

ML Week 7

Number of participants: 73



1. Bagging

3 correct answers
out of 14 respondents



Uses bootstrap data sets to fit many prediction models.



12 votes

If the number of trees $B = 1$, then the training error is 0.



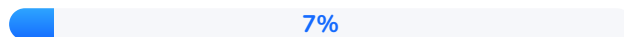
5 votes

Would also work if all trees only have one split.



8 votes

If B is too large, the test error will increase.



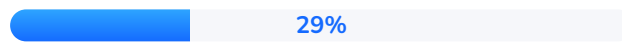
1 vote



2. Random Forest

3 correct answers
out of 14 respondents

Generally has a smaller bias than bagging.



29%

4 votes



Generally has a smaller variance than bagging.



43%

6 votes



Contains bagging as a special case.



57%

8 votes

Taking too many trees will overfit the data.



14%

2 votes



3. Bagging vs Random Forests

4 correct answers
out of 12 respondents

Only random forests allow for out-of-bag estimation.



2 votes

Both bagging and random forests have the tuning parameter m .



2 votes



Trees tend to be more similar to one another with bagging.



9 votes

In random forests, each tree considers a fixed random subset of features for all its splits.



5 votes



4. Which of the following(s) is true?

11 correct answers
out of 12 respondents

In random forests, you decrease the test error by increasing the correlation between the trees.

0%

0 votes

Prediction from a random forest are random due to their feature subset selection and bootstrap resampling.



92%

11 votes

In bagging, each bootstrapped tree has the same number of leaves as the number of observations in the original training data.

8%

1 vote

In random forest and bagging, changing the number of trees is a bias-variance trade-off.

0%

0 votes